MINING DEALER BEST PRACTICE

Spreader Bar Protection

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The dragline uses hoist chains to lift the bucket up and lower it down. Hoist chains are on both sides of the bucket. A spreader bar is between the upper hoist chain and lower hoist chain and it is used to prevent the hoist chains rubbing on the bucket. See below picture.

Figure 1 Spread bar with rubber protection

The dimensions of a new spreader bar will vary depending upon the size of the bucket.

Figure 2 New spreader bar

2.0 Best Practice Description

For initial operation of the dragline, especially for new operators, the service life of the spreader bar is very short, about two months. Sometimes even shorter to about a few hours depending on the operator's skills. Once it was broken, maintenance people must rush to the site, a crane must be used to restore the dragline back to operation as soon as possible, even at night shift.

Figure 3 Broken spreader bar


Figure 4 Typical Failures

Most of spreader bars were broken by hitting the top of the bucket. The key to prolong the service life of the spreader bar is to prevent it from hitting on the bucket, which seems impossible. Given they cannot completely avoid the spreader bar hitting against bucket, a method to reduce the hitting force was created. Used rubber tires are put on the spreader bar to reduce direct impact to spreader bar. It has been proven this works effectively. See Pic 5 and Pic 6 below.

Figure 5 Spreader bar without rubber protection

Figure 6 Spreader bar with rubber protection

3.0 Implementation Steps

NOTICE
Dispose of all rubber tires according to local regulations and mandates.

WARNING!
IT IS ESSENTIAL THAT ALL APPROPRIATE SAFETY MEASURES SUCH AS THOSE SET FORTH IN THE APPLICABLE MACHINE MAITENANCE MANUAL ARE FOLLOWED UP ACCORDING THE PERFORMACE OF THIS WORK.

Find and collect the old useless rubber tires, make sure the inner hole diameter is over 20 inches and the outer diameter is roughly the same. Put them on the spreader bar one by one closely.

Depending on the tube size and length of the spreader bar, the quantity and size of the rubber tires used on the spreader bar will vary accordingly.

EDITORS NOTE: Please note the extra weight added to the boom by this practice. Follow appropriate safety measures to ensure boom is not over its rated capacity. This may require not filling buckets to capacity.

4.0 Benefits

The exact benefits per year is difficult to calculate, but according to customer's statistics:

Money Saved

- Repair Cost:
  - Over $800K USD per year
  - It includes the machine down time, employee's salary, the auxiliary equipment and tools, such as cranes, trucks, all materials consumed, such as welding rods, oxygen and acetylene, angle grinders and so on.
- Increased uptime
  - It also increased the dragline uptime and made the customer satisfied and increased loyalty.

5.0 Resources Required

- Old useless rubber tires
- Trucks
- Cranes
- General tools

6.0 Supporting Attachments / References

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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